Akshat Kankani

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Overview

Computer Science (AI) undergraduate experienced in architecting scalable full-stack applications and deploying end-to-end machine learning pipelines. Strong foundation in software engineering, with expertise in building robust REST APIs, web interfaces, and distributed backends using modern frameworks (Next.js, FastAPI, PostgreSQL). Skilled in applied AI and data engineering—specializing in NLP (Transformers, RAG architectures), automated data preprocessing, and model evaluation to transform raw, real-world data into production-ready intelligence.

Education

Manipal Institute of Technology — Bengaluru, KA
Bachelor of Technology in Computer Science (Artificial Intelligence) — *July 2023 – May 2027*

- CGPA: 8.01

Projects

BiasScope | *Python, Hugging Face Transformers, PostgreSQL, FastAPI, Next.js* — Live Demo | GitHub

- Architected a full-stack news intelligence platform that aggregates global media, detects political bias, and enables cross-examination via a RAG-based chat interface, built on a distributed microservices architecture.
- Built an asynchronous ingestion pipeline (NewsAPI + headless scraping) to process 50+ sources with deduplication and anti-bot fallbacks, achieving 90% data quality across noisy real-world datasets.
- Developed and deployed NLP pipelines using Hugging Face Transformers (DistilBERT, PoliticalBiasBERT) for bias and sentiment classification, enabling scalable, low-latency inference on live news streams.
- Implemented a Retrieval-Augmented Generation (RAG) system using LLaMA-3-8B with strict context grounding, reducing hallucinations and ensuring fact-based responses from source articles.
- Engineered a secure Next.js + PostgreSQL backend (Prisma ORM, Better-Auth) supporting persistent sessions, historical analytics, and automated PDF intelligence report generation.

DataScope | *Python, Pandas, scikit-learn, PostgreSQL, FastAPI* — Live Demo | GitHub

- Architected a full-stack ML validation platform with automated pipelines for consensus-based outlier detection, data leakage checks, and comprehensive dataset drift analysis.
- Engineered a Scikit-learn evaluation engine utilizing dynamic cross-validation to quantify the causal impact of data-cleaning techniques on accuracy and R^2 scores.
- Implemented intelligent preprocessing heuristics that leverage SHAP values to visualize feature importance and optimize dataset quality for model training.
- Designed an interactive analytics dashboard featuring an automated technical glossary, EDA plots, converting complex statistical diagnostics into actionable, interpretable visual insights.
- Deployed an end-to-end architecture using Next.js, FastAPI, and PostgreSQL to handle secure dataset uploads, real-time ML processing, and automated reporting.

Low-Latency Matching Engine | *Python, Data Structures, Algorithm Design, Micro-Optimization, Benchmarking*

- Engineered a production-grade limit order book matching engine in Python, implementing price-time priority (FIFO) with O(1) order insertion and cancellation using optimized data structures.
- Designed a hybrid data structure (hash map + sorted price levels + deque) to achieve efficient trade matching (O(log n) price lookup, O(1) per fill), replicating real-world exchange mechanics.
- Implemented support for LIMIT and MARKET orders with partial fills, multi-level matching, and lazy cancellation handling, ensuring correctness under high-frequency trading scenarios.
- Developed a benchmarking suite simulating realistic workloads (resting, crossing, mixed) achieving 377K orders/sec throughput and 10µs median latency under matching conditions.
- Built a comprehensive test suite validating edge cases including price-time priority, multi-level fills, order cancellation, and cross-symbol isolation.

Technical Skills

Languages: Python, C, C++, Go
Databases: MySQL, PostgreSQL
Libraries/Frameworks: NumPy, pandas, Matplotlib, seaborn, scikit-learn, TensorFLow, Keras, PyTorch, FastAPI
Developer Tools: Git, GitHub, VS Code, Jupyter Notebooks, Google Colab, Postman, PowerBI, Vercel
Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, OOP, Machine Learning